

Lec 49

Bubble sort

quick sort merge sort

Lec_49: selection Sort

Array $\Rightarrow$ [10, 5, 9, 3, 21, 1, 20] $\rightarrow$ Bubble sort

$O(n^2)$ in worst case.

$O(n)$ in best case with modified Bubble sort.

$\rightarrow$ Bubble sort gives $O(n)$

if array is sorted or almost sorted (But need to modify)

Count no. of swaps in each iteration $\leftarrow$ if 0 return or break

Ex:

$\{10, 2, 32, 1, 21, 20, 3\}$

↓

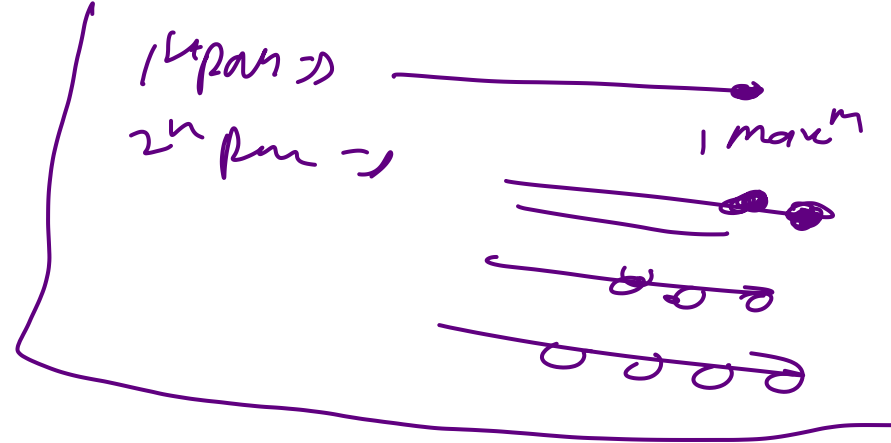
after sorting $a[1, 2, 3, 10, 20, 21, 32]$

$\begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 \\ 10 & 2 & 32 & 1 & 21 & 20 & 3 \end{matrix}$

index of
smallest
element at
each iteration

$\uparrow \Rightarrow$ compare $a[i]$ and $a[\text{minindex}]$
if $a[i] < a[\text{minindex}]$
 $\{ \text{minindex} = i \}$

after first iteration $\text{minindex} = 3$
after jth loop, swap($a[i]$, $a[\text{minindex}]$)



for($l=0$; $l < n$; $l++$)

{ minindex = i

for($j = i+1$; $j < n$; $j++$)

{ if($a[j]$ < $a[\text{minindex}]$)

{ minindex = j

}
}
swap($a[i]$, $a[\text{minindex}]$)

Time complexity of Selection Sort is $O(n^2)$ in each

TC = $O(n^2)$

no. of swaps = $O(n)$

1 head
can

$\begin{bmatrix} 10 & 5 & 4 & 20 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$
min

$\begin{bmatrix} 10 & 5 & 4 & 20 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$
min = 0 at $i=1, j=2$

$\Rightarrow n, n-1, n-2, \dots, 1$

$\Rightarrow n + n-1 + n-2 + \dots + 1$
 $\Rightarrow \frac{n(n+1)}{2} = O(n^2)$

- $T_c = O(n^2)$ each row.
 - no. of swaps = $O(n)$ in each row.
 - not stable (insertion, quicksort, selection sort)
 - in place
- Insertion

